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REGULAR ARTICLE

Is it possible to manipulate root anchorage in young trees?

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Abstract The optimal root system architecture for increased tree anchorage has not yet been determined and in particular, the role of the tap root remains elusive. In Maritime pine (*Pinus pinaster* Ait.), tap roots may play an important role in anchoring young trees, but in adult trees, their growth is often impeded by the presence of a hard pan layer in the soil and the tap root becomes a minor component of tree anchorage. To understand better the role of the tap root in young trees, we grew cuttings (no tap root present) and seedlings where the tap root had (–) or had not (+) been pruned, in the field for 7 years. The force (F) necessary to deflect the stem sideways was then measured and divided by stem cross-sectional area (CSA), giving a parameter analogous to stress during bending. Root systems were extracted and root architecture and wood mechanical properties (density and longitudinal modulus of elasticity, E_L) determined. In seedlings (–) tap roots, new roots had regenerated where the tap root had been pruned, whereas in cuttings, one or two lateral roots had grown downwards and acted as tap roots. Cuttings had significantly less lateral roots than the other treatments, but those near the soil surface were 14% and 23% thicker than plants (+) and (–) tap roots, respectively. Cuttings were smaller than seedlings, but were not relatively less resistant to stem deflection, probably because the thicker lateral roots compensated for their lower number. Apart from stem volume which was greater in trees (+) tap roots, no significant differences with regard to size or any root system variable were found in plants (–) or (+) tap roots. In all treatments, lateral roots were structurally reinforced through extra growth along the direction of the prevailing wind, which also improved tap root anchorage. Predictors of log F/CSA differed

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depending on treatment: in trees (–) tap roots, a combination of the predictors stem taper and %volume allocated to deep roots was highly regressed with log F/CSA ($R^2 = 0.83$), unlike plants (+) tap roots where the combined predictors of lateral root number and root depth were best regressed with log F/CSA ($R^2 = 0.80$). In cuttings, no clear relationships between log F/CSA and any parameter could be found. Wood density and E_L did not differ between roots, but did diminish with increasing distance from the stem in lateral roots. E_L was significantly lower in lateral roots from cuttings. Results showed that nursery techniques influence plant development but that the architectural pattern of Maritime pine root systems is stable, developing a sinker root system even when grown from cuttings. Anchorage is affected but the consequences for the long-term are still not known. Numerical modelling may be the only viable method to investigate the function that each root plays in adult tree anchorage.

Keywords *Pinus pinaster* Ait · Root pruning · Tree stability · Biomechanics · Root architecture · Stem lean, Toppling

Introduction

Research into tree stability is now well established within northern Europe (Coutts and Grace 1995; Gardiner and Quine 2000; Ruck et al. 2003; Cucchi and Bert 2003; Nicoll et al. 2005; Danjon et al. 2005), especially since the 1999 storms, Lothar and Martin, which resulted in huge losses of timber. Decision support systems also now exist, which incorporate biomechanical models and can provide a risk factor for a given stand depending on tree stability characteristics and silviculture (Gardiner et al. 2000; Cucchi et al. 2005). However, although knowledge on root anchorage is increasing steadily (Coutts 1986; Mickovski and Ennos 2003; Cucchi et al. 2004; Danjon et al. 2005; Nicoll et al. 2005), most information is not yet applied in a practical manner. Certain tree stability characteristics have already been identified e.g. the importance of symmetrical lateral roots (Coutts et al. 1999), and shallow, windward lateral roots (Coutts 1986; Stokes et al. 1995) and others still need to be elucidated e.g. the role of the tap root (Hintikka 1972). Danjon et al. (2005) showed that in mature Maritime pine (*Pinus pinaster* Ait.), anchorage was provided by a large tap root, horizontal surface roots and numerous sinkers and deep roots, which composed a rigid 'cage,' encompassing a large mass of soil and which was guyed by long horizontal surface roots. Once understood, such root system characteristics contributing to anchorage may be manipulated either by the tree breeder or the forester, or used to develop root anchorage models (Dupuy et al. 2005a, b) which could then be incorporated into decision support systems (Mickovski et al. 2005).

Maritime pine is a major timber species in southern Europe, and is highly prone to trunk flexuosity (Danjon 1994; Fourcaud et al. 2003) and to a lesser extent, mechanical instability, or toppling, during wind storms (Cucchi and Bert 2003; Cucchi et al. 2004). When subjected to mechanical stress through e.g. wind loading, adaptive growth occurs quickly, resulting in increased resistance to overturning (Stokes 1999; Berthier and Stokes 2006), Maritime pine could become both straighter and more mechanically stable, if managed with these factors as a priority. Not only are root wood mechanical properties influenced by mechanical stress but individual root shape can also be altered which results in increased rigidity (Stokes et al. 1998). Root systems can be manipulated fairly easily during the juvenile stage of tree growth e.g. by removing roots through pruning, a standard practice for certain species during transplanting (Harrington and Howell 1998; Jinks and Kerr 1999; Krasowski and Owens 2000). Thus, the branching pattern of the remaining root system can be altered (Burschel and Stimm 1993; Girouard 1995). Structural roots are laid down early on in the life of a tree (Coutts and Lewis 1983) therefore, if the optimal morphology of a root system was known, with regard to overturning resistance, nursery techniques could be adapted to encourage the formation of this type of root system (Burdett et al. 1986). The morphology and topology of the root system will continue to change throughout the life of the tree, depending largely on soil conditions.

Nevertheless, the basic geometric structure laid down close to the stem usually remains in place, at least throughout the short life of a temperate forest timber species (Coutts and Lewis 1983).

One million hectares of Maritime pine are grown in the south-west of France on two different types of soil: dry and wet Landes. The former consists of a deep water table and broken or negligible hard pan at depths of about 1.0 m (Cucchi et al. 2004). In wet Lande however, a discontinuous hard pan can be found at depths of 0.4–1.2 m and the water table is shallow, therefore inhibiting vertical root growth (Danjon et al. 2005). Both hard pan and water table depth are strongly related to the microtopography (Danjon et al. 2005). When the rooting depth is unlimited, Maritime pine forms a long, thick central tap root, with smaller, shallow lateral roots to which a large number of vertical sinker roots are attached. However, if a hard pan is present, tap root and vertical sinker root length is inhibited, and roots fuse together on the underside of the soil-root plate (Danjon et al. 1999a, b, 2005), resulting in a smooth surface with little adhesion between the roots and soil. As the tap root cannot penetrate the hard pan horizon found in wet Lande, a short, thick tap root develops in adult trees, which cannot ensure the same anchorage functions as if this large, central root had been able to grow to its full extent (Ennos 1994; Crook and Ennos 1997). Therefore, in adult trees, anchorage functions are largely taken over by sinker roots and the volume allocated to the zone of rapid taper (ZRT), i.e. the highly tapered zone near the trunk in lateral roots (Danjon et al. 2005). Hence, it could be considered that the resources used by the tree in developing the tap root are unnecessary once the juvenile stage has passed.

Although the tap root plays a secondary role in adult Maritime pine trees on shallow soils with regard to anchorage strength, in young trees the tap root may well serve an important role in preventing the juvenile tree from being uprooted during strong storms (Auberlinder 1982, Burdett et al. 1986; South et al. 2001). As the tap root in young trees is relatively longer with regard to shoot height and volume compared to older trees (Danjon et al. 1999a), the tap root acts like a pole in the soil, and the lateral roots as guy ropes holding the trunk in place (Ennos 1994; Danjon et al. 2005). If the tap root was missing or reduced in size, anchorage would be less efficient, and permanent stem deflection or toppling would occur during a storm, if not root failure (Balneaves and De La Mare 1989). Any stem deflection occurring during this juvenile stage would also aggravate trunk flexuosity, a serious problem in Maritime pine.

In order to elucidate the role of the tap root in the anchorage of young Maritime pine, one year old seedlings and cuttings were planted in the field. Seedlings generally possess their original tap root whereas cuttings do not, and only adventitious lateral roots are present (Sasse and Sands 1997). Some seedlings had their tap root pruned (undercutting) and others were left untouched. Stem straightness was determined and mechanical tests were carried out on saplings after seven years' growth, to test if some trees were more resistant to bending than others. The force necessary to deflect stems sideways in four directions around the tree was measured, to determine if trees had adaptively responded to loading by the prevailing winds. Trees were then uprooted and root architecture measured. Mechanical properties were also measured on lateral and tap roots, to ascertain if differences in tree anchorage were due to root wood elasticity or root system architecture.

Materials and methods

Plant material and site conditions

One year old seedlings and cuttings of Maritime pine were planted in a 500 m^2 plot at the Domaine de l'Hermitage, Pierroton, Gironde (58 m altitude; latitude 44°44′ N, longitude 0°46′ W), in May 1995. Soil is typical of wet Lande, a completely flat sandy podzol with a discontinuous hard pan horizon at a depth of 0.5–1.0 m. Annual precipitation is 990 mm (Cucchi et al. 2004) and the prevailing wind direction is from the north-west (weather station located at 2 kms from our site). Young trees were planted in

rows (aligned from east to west) with a spacing of 1 × 4 m. 240 plants were transplanted; 120 bare root cuttings with a mean height of 0.18 ± 0.04 m and 120 seedlings with a mean height of 0.30 ± 0.1 m (data are means ± standard error). The latter had been sown in 50 × 100 mm paper pots which were not removed, as these degrade over time. The tap root was lignified with a diameter of 2–3 mm and extruded from the base of the paper pot for most seedlings. The tap root was removed from half the seedlings by cutting it with scissors at the base of the pot. Trees were planted as single tree plots, i.e. in the order: cutting, seedling with (+) tap root, cutting, seedling without (–) tap root, cutting etc. Stem length and diameter at breast height (DBH) for all trees were measured in November 2001 (Table 1). Stem extension for the years 1996, 1999, 2000 and 2001 was also measured, by measuring the distance between each major whorl of branches (Danjon 1994, one major whorl corresponds to one year in Maritime pine). Trees with high taper are often considered more resistant to wind loading (Telewski 1995) therefore stem taper was calculated using stem length/DBH in 2001. 76 trees were harvested in March 2002 (39 cuttings, 19 (+) tap root and 19 (–) tap root) for measurements of root architecture. These trees were chosen from the inside of the stand (to avoid edge effects) and were considered healthy without any obvious defects e.g. stem forks or pathogen attack.

Measurements of stem straightness

Stem basal sweep or lean can be interpreted as a sign of poor anchorage in Maritime pine (Maugé 1987). To test if stem straightness differed between treatments after 8 years of growth, during which time the saplings had been subjected to the wind storm Martin in December 1999, measurements of stem lean were made in March 2002. The distance between the trunk at breast height (BH) and a pole placed upright at the stem base was measured, along with the azimuth of lean. The distance was converted into an angle of lean using simple trigonometry (Stokes and Berthier 2000).

Resistance to mechanical deflection

Forty-two trees were chosen for mechanical testing from the 76 already selected. In November 2001, DBH and D at 0.1 m above ground level were measured in two directions (N to S and E to W) and the mean calculated for each tree (Table 1). Stem cross-sectional area (CSA) was then calculated at both these heights along the stem. Mean stem length was 2.59 ± 0.99 m and diameter at 0.1 m above ground level was 0.06 ± 0.01 m (Table 1). The resistance of these trees to horizontal deflection was measured using a method similar to that of Mickovski and Ennos (2002). In March 2002, trees were tested by applying a force to the trunk and pulling it

Table 1 Mean shoot growth parameters ± standard error for the 76 cuttings and trees with (+) or without (–) tap roots in 2001, unless stated otherwise

Growth parameter	Treatment			*P*
	Cutting	(–) Tap root	(+) Tap root	
Stem length in 1995 (m)	0.18 ± 0.04^{a}	0.30 ± 0.02^{b}	0.29 ± 0.02^{b}	**<0.001**
Stem length in 2001 (m)	2.5 ± 0.7^{a}	2.8 ± 1.3^{b}	3.1 ± 1.5^{b}	**<0.001**
Stem length of plants used for mechanical analysis (m)	2.4 ± 0.1^{a}	$2.7 \pm 0.2^{a,b}$	2.9 ± 0.2^{b}	0.128
DBH (m)	0.06 ± 0.01^{a}	$0.07 \pm 0.01^{a,b}$	0.08 ± 0.01^{b}	**<0.001**
DBH of plants used for mechanical analysis (m)	0.07 ± 0.01^{a}	$0.08 \pm 0.01^{a,b}$	0.10 ± 0.01^{b}	**0.038**
D at 0.1 m of plants used for mechanical analysis (m)	0.11 ± 0.01^{a}	0.14 ± 0.01^{b}	0.15 ± 0.02^{b}	**0.004**
Stem volume (m^3)	0.015 ± 0.004^{a}	0.016 ± 0.008^{b}	0.018 ± 0.009^{c}	**<0.001**
Stem lean (°)	9.9 ± 0.7	11.7 ± 1.0	12.0 ± 1.2	0.128
Stem taper	17.8 ± 0.7^{a}	31.5 ± 3.1^{b}	28.0 ± 1.3^{b}	**<0.001**

Where letters in superscript differ, differences are significant between treatments using a Fisher LSD test ($P < 0.05$). In the Probability (*P*) column, *P* values (significant *P* values are in bold) are given for the comparison of all three treatments together using ANOVA (without DBH as covariate)

sideways, but without causing any permanent stem displacement. Young pine trunks are very flexible and can be severely deflected before root failure occurs. Therefore, as root anchorage only was to be measured, the force was applied just above ground level, also avoiding any supplementary effects due to differences in stem mechanical properties between trees. To determine if resistance to deflection was greater in any one direction, each stem was pulled sideways in four directions around the tree with regard to the prevailing wind direction (NW) at the site i.e. towards NW, SW, SE, NE. The order of testing was random and the amount of force used was not enough to cause plastic failure to the tree through e.g. root slippage or stem breakage. A strain transducer (HBM DD1) was positioned on the trunk at a height of 0.1 m above ground level. Two pins attached to the transducer were pushed into the wood thus fixing it to the tree (Stokes et al. 1998). The transducer was connected to a strain indicator (Kyowa SDM10). A strain transducer was used for ease and speed, instead of a displacement transducer and it was not thought that differences in wood properties between trees would be sufficient to bias results. A 3 m long sling was then knotted around the trunk, 0.02 m above the transducer. A 20 kN capacity load cell (Scaime S.A., K25H) was hooked onto the end of the sling and was connected to a datalogger (Almemo 2290-8, Ahlborn), which measured force applied every second. A second hook was fixed to the other end of the datalogger and force was applied manually at a rate of approximately 10 μdeformations s^{-1}, by pulling on this hook until a displacement of 100 μdeformations had been reached on the strain indicator. During the pulling tests, the sling was kept horizontal and the total distance between the tree stem and the point of application of force (F) was 3 m. The tree was then allowed to return to the vertical position and the same procedure was carried out again, until the horizontal deflection had been measured at four points around the tree. To compare differences between trees, which were variable with regard to height and diameter, the F applied was divided by stem CSA at a height of 0.10 m. Although not a direct calculation of the stress applied, because F was applied in bending, F/CSA can be considered as analogous to stress during bending.

Measurements of root growth

In March 2002, ground cover was removed from around each of the 76 trees already selected, using a small spade. The earth was then loosened with an air knife (Rizzo and Gross 2000) and trees were uprooted using a mechanical digger. Care was taken to note the direction to north on each tree. Trees were stored in a nearby lake (water temperature: 5–10°C) to avoid dehydration until root measurements were carried out (within several weeks of uprooting).

Stems were removed from trees at a height of 0.5 m. This remaining stem was then inserted upside-down in a plastic drain-pipe so that the root system could be measured more easily. Using the method developed by Danjon et al. (1999a), the spatial coordinates of each root were digitized using a Polhemus Fastrak low magnetic field digitizer (Polhemus, Colchester, VT, USA) and DIPLAMI software (Sinoquet et al. 1997). Root diameter was measured simultaneously using a pair of vernier callipers. Digitizing consisted of simultaneously recording the topology and XYZ position of measurement points (Danjon et al. 1999a, b). All roots with a diameter at the origin >2 mm were measured. In all three populations, the largest vertical root originating from the stump was coded as a tap root and had an order 1.

The characteristics of trees, axes and root segments were computed in the same way as that described in Danjon et al. (1999a, b, 2005) and Tamasi et al. (2005), using the AMAPmod software[1] (Godin and Guédon 1999; Godin et al. 1999) and R software under the Linux Mandrake system.

Specific root surface (SRS) was calculated as root surface area divided by root volume and specific root length (SRL) was calculated as root length divided by root volume (SRL). Relative

[1] Freely available for Linux and Windows platforms: www.cirad.fr and ftp.cirad.fr/pub/amap/AMAPmod—AMAPmod and R functions for computation of the root characteristics described here are available from F. Danjon.

root number was determined as the total number of roots divided by the diameter at the stem base. The mean number of lateral roots emerging from the central axis was separated into those on the stump and those below the stump. The allocation of resources to the root system was calculated as in Danjon et al. (1999a, b; 2005) as a 'root partitioning coefficient' (RPC; i.e. root volume/(root volume + stump volume + stem volume)). According to Danjon et al. (2005), the root segments were classified into eight compartments i.e. (1) stump, (2) tap root, (3) ZRT, (4) horizontal shallow roots beyond ZRT, (5) sinkers, (6) deep roots, (7) intermediate depth roots and (8) oblique roots. The limits between horizontal, oblique and sinker roots were 30° and 60° towards the soil surface respectively. The initial branching point was used to classify the individual roots as a function of their depth. The distance to the soil surface limits between shallow, intermediate depth and deep roots were set to 0.23 and 0.62 m, respectively, which corresponded to 25% and 66% of the maximal rooting depth. We then also calculated the total volume of all root segments in each of these layers, regardless of whether they were stump, lateral, sinker or tap root segments. A certain number of tap roots were inclined. Therefore, in the compartment and circular distribution computations, radial distance and azimuth of a segment were not computed relative to the collar (X,Y) position, but relative to the end of the tap root segment which bears the corresponding root arborescence. Sinkers were considered to be beneath the ZRT when the relative radial distance of their initial branching point was less than three times the collar diameter. The ZRT included the last segment of each axis for which the taper from axis origin was larger than 2.5% per 0.01 m and all segments located between the axis origin and this last segment (see Danjon et al. 2005). The proportion of root volume allocated to each of the compartments was then calculated.

To determine if differences in root system asymmetry occurred with regard to the prevailing wind direction, data were sorted into four sectors: windward (WW), leeward (LW) and perpendicular to the wind direction (PP), which also corresponded to the directions in which the trees were pulled. The total %volume allocated to each sector, along with the %volume allocated to the ZRT, horizontal roots beyond the ZRT, sinker roots, intermediate and oblique roots (combined) and deep roots was calculated along with the % root number and length in each quadrant. For the PP quadrants, data used were the means from both quadrants, as it is considered that no differences should occur between these two quadrants (Stokes et al. 1995).

Mechanical properties of individual roots

Measurements of longitudinal modulus of elasticity (E_L) and wood density were carried out on lateral and tap roots selected from the trees whose root systems had been measured. In order for measurements to be performed correctly, roots had to be reasonably straight and >10 mm in diameter. The bark was removed from 26 tap roots and 66 lateral roots. Roots were then cut into 0.1 m long sections but if a fork or daughter branch was present, the section was rejected as the presence of such anomalies would significantly influence E_L. As wood moisture content affects E_L, it was necessary to dry all samples to a constant weight. Using specific equations (e.g. Guitard 1987), the E_L of fresh wood can be determined if desired. Therefore, roots were dried in a chamber with 60% humidity. Each sample was weighed regularly until all roots reached a constant weight. The position of each sample with regard to type, azimuth, depth and distance from the stem was noted, as well as diameter measured in the middle of the sample (n = 125).

E_L was measured using an Ultrasound meter (Lucchi Minipalm Meter; www.lucchicremona.com):

$$E_L = \rho c^2 \quad (1)$$

where ρ is wood density and c is the speed of the ultrasound wave. c was calculated by dividing the sample length by the time taken for the ultrasound wave to travel through the sample. ρ was calculated by weighing each sample before submerging it in a beaker of distilled water placed on a balance and then measuring the increase in

mass. Assuming that 1,000 mm^3 of water = 1 g mass, we used Eq. 2 to determine ρ:

$$\rho = \frac{P}{V} \quad (2)$$

where P is the mass of the sample before submersion in water and V is the volume of water displaced.

Statistical analysis

The mean azimuthal direction (and associated confidence intervals, CI, at 95%) in which trees were leaning was calculated using circular statistics methodology (Batschelet 1981). Normality of data for each treatment was tested using an Anderson-Darling test (data were normally distributed when $P > 0.05$). With the exception of the root mechanical data, distributions were not always normal, therefore non-proportional data were log-transformed before analysis. All proportional data were arcsine square root transformed before analysis. Analysis of variance was used to test differences between treatments with regard to mechanical resistance of the trees in different directions and the mechanical properties of roots from different zones around the tree, using log DBH as a covariate where appropriate. The mean *F*/stem CSA at 0.1 m height for each tree was then calculated using data from the four directions in which trees were deflected and used for further analyses. Stepwise regression was carried out between mean log *F*/stem CSA at 0.1 m height and stem and root architectural variables, to test which predictor, or combination of predictors (Bailey et al. 2002; Dupuy et al. 2005a; Stokes et al. 2007), were the best indicators of root anchorage for all treatments combined and then for each individual treatment. Where the coefficient of determination (R^2) was low, i.e. <0.50, results are not shown, even if the probability (P) was ≤0.05. Differences between treatments were determined with regard to these relationships, using stem and root system variables as covariates. Fisher Least Significant Difference (LSD) tests were used to compare significant differences between treatments and directions.

Results

Plant performance

Cuttings were significantly shorter than seedlings in 1995 and 2001 (Table 1). No differences in stem extension were found 1 year after transplanting, but 3 years later, stem extension was significantly greater in plants (+) tap roots compared to cuttings (Fig. 1). In the years 2000 and 2001, no significant differences in stem extension were found between any treatment (Fig. 1) but cuttings were still significantly shorter than both other treatments in 2001 (Table 1). Plants (+) tap roots had a significantly larger DBH in 2001 than cuttings only (Table 1). Stem volume was significantly greater in plants (+) tap roots compared to both other treatments and significantly greater in plants (–) tap roots with regard to cuttings (Table 1). Stem taper was significantly less in cuttings only (Table 1). No significant differences in stem length, extension, taper, diameter at 0.1 m or DBH were found between plants (+) or (–) tap roots (Table 1).

Measurements of stem straightness

Most trees were found to lean slightly (10.90 ± 0.53° for all trees) along the direction of the prevailing wind (mean azimuth = 107°, (CI = 71°, 143°)). Cuttings were slightly straighter than seedlings, but this difference was not significant (Table 1).

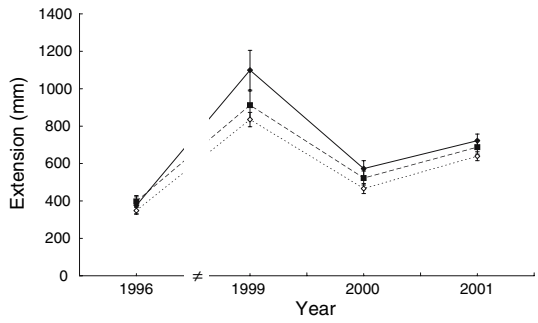


Fig. 1 Stem extension for seedlings (+) tap roots (black diamonds, solid line), (–) tap roots (black squares, dashed line) and cuttings (white diamonds, dotted line) from 1996 to 2001. Means are ± standard error

Resistance to mechanical deflection

Mean F was significantly greater in trees (+) and (–) tap roots compared to cuttings, using D at 0.1 m as a covariate ($F_{2,41} = 5.56$, $P = 0.008$). No significant differences in F were found between trees (+) or (–) tap roots, nor with regard to azimuth. When the mean F required to pull trees sideways was divided by CSA at 0.1 m height (161.4 ± 10.4 Nm^{-2} for all trees), no significant differences were found between any treatment, even when shoot parameters measured were used as covariates. The mean log F/CSA necessary to displace all stems by a given distance decreased significantly with increasing stem volume ($y = -1.97x + 5.30$, $R^2 = 0.46$, $P < 0.001$) and DBH ($y = -1.16x + 3.65$, $R^2 = 0.41$, $P < 0.001$) although R^2 was too low to be considered of importance.

Measurements of root growth

When the young trees were excavated and root systems measured, no deformations of the root systems due to the presence of paperpot containers left around the seedlings' root systems were observed. Mean total root volume for all trees was 0.013 ± 0.001 m^3 but was significantly smaller in cuttings compared to other treatments (Table 2). The RPC was not significantly different between treatments. SRS and SRL were significantly smaller in cuttings compared to plants (–) tap roots only and maximal radial distance was shorter than in plants (+) tap roots (Table 2). The mean number of lateral roots emerging from the stump was significantly less in cuttings (8.8 ± 0.6) compared to both other treatments (20.5 ± 1.2 for (–) tap roots and 19.6 ± 1.4 for (+) tap roots, Table 2). The total relative root number was also significantly lower in cuttings compared to seedlings whereas the inter-lateral length of lateral roots along the tap root was significantly longer (Table 2). However, the mean basal diameter of lateral roots emerging from the stump was significantly greater in cuttings (17.0 ± 0.1 mm) compared to plants both (–) (13.1 ± 0.1 mm) and (+) (14.7 ± 0.1) tap roots (Table 2). No significant differences between treatments were found with regard to maximal depth or tap root length (Table 2).

In seedlings where tap roots had been pruned, one or two tap roots had regenerated where the original tap root had been cut. In cuttings, it appeared that one of the lateral roots had grown downwards to 'replace' the tap root (Fig. 2). When root systems were divided into several different compartments with regard to volume, significant differences were found between cuttings and other treatments. Cuttings had significantly less sinker root volume both throughout the whole root system and also in the region beneath the ZRT (Table 2). However, cuttings possessed a significantly higher volume of oblique roots compared to plants (–) tap roots only (Table 2).

A significant or nearly significant difference in the total %root volume allocation with regard to wind direction was found in the three treatments (Table 3). The WW and LW quadrants both showed approximately 20% more reinforcement, at the expense of the PP sector, except for the trees (–) tap roots which showed no WW reinforcement (Table 3). The same pattern was also observed for root length, with the sinkers and the shallow roots beyond the ZRT possessing 20–40% longer roots on the leeward side (Table 3). Conversely, only the plants (–) tap roots showed a circular heterogeneity with regard to root number, with 25% extra reinforcement on the LW side (Table 3). In sinker roots of cuttings, almost twice as much volume was allocated to LW roots compared to the WW quadrant (Table 3). No circular heterogeneity was found in the intermediate depth, deep or oblique root compartments.

Predictors of root anchorage

When all trees were considered together, regressions of log F/CSA and root system parameters showed that the best predictors of anchorage were not combined, but individual predictors. The volume allocated to roots in the shallow ($y = 0.95x + 2.08$, $R^2 = 0.43$, $P < 0.001$) and intermediate depth ($y = -1.01x + 3.60$, $R^2 = 0.40$, $P < 0.001$) compartments only were significantly regressed with log F/CSA, but R^2 was too low to indicate straightforward relationships. No

Table 2 Mean root growth parameters ± standard error for the 76 cuttings and trees with (+) or without (–) tap roots

Growth parameter	Treatment			*P*
	Cutting	(–) Tap root	(+) Tap root	
RPC (%)	0.20 ± 0.08	0.20 ± 0.09	0.18 ± 0.11	0.209
Total root volume (m^3)	0.009 ± 0.001^a	0.014 ± 0.002^b	0.018 ± 0.003^b	**0.003**
Total root length (m)	9.7 ± 1.0^a	19.9 ± 1.0^b	22.7 ± 2.0^b	**<0.001**
Specific root surface (m^{-2})	360 ± 20^a	430 ± 30^b	400 ± 30^{a,b}	**<0.001**
Specific root length (m^{-1})	16000 ± 2000^a	24000 ± 3000^b	20000 ± 3000^{a,b}	**<0.001**
Maximal radial distance (m)	1.4 ± 0.8^a	1.5 ± 0.8^{a,b}	1.6 ± 1.0^b	0.124
Maximal depth (m)	0.7 ± 0.2	0.7 ± 0.3	0.7 ± 0.2	0.216
Tap root length (m)	0.8 ± 0.3	0.8 ± 0.5	0.8 ± 0.4	0.792
Relative root number (number/DBH (m))	389.9 ± 16.2^a	578.0 ± 39.1^b	564.7 ± 22.7^b	**<0.001**
Mean number of lateral roots on stump	8.8 ± 0.6^a	20.5 ± 1.2^b	19.6 ± 1.4^b	**<0.001**
Mean basal diameter of lateral roots on stump (mm)	17.0 ± 0.1^a	13.1 ± 0.1^b	14.7 ± 0.1^{a,b}	**0.031**
Mean inter-lateral length along tap root (mm)	51.6 ± 3.9^a	29.3 ± 3.1^b	27.1 ± 2.1^b	**<0.001**
Proportion of stump volume (%)	49.3 ± 1.6	49.6 ± 1.3	46.5 ± 1.1	0.128
Root wood density (kg/m^3)	422.8 ± 7.7^a	454.9 ± 19.5^b	456.7 ± 7.6^b	**0.050**
Shallow soil layer (all root segments) (%)	69.0 ± 2.9	65.6 ± 4.7	61.8 ± 2.8	0.33
Intermediate soil layer (all root segments) (%)	29.9 ± 2.7	31.7 ± 4.3	35.9 ± 2.6	0.41
Deep soil layer (all root segments) (%)	1.1 ± 0.3	2.1 ± 0.6	1.9 ± 0.5	0.12
Proportion of root volume (%) between different compartments (with stump removed)				
Horizontal shallow segments within ZRT	17.1 ± 2.0	15.8 ± 2.0	15.8 ± 1.6	0.86
Intermediate roots	6.2 ± 1.8	5.1 ± 2.2	4.3 ± 1.1	0.76
Oblique roots	12.4 ± 2.4^a	4.6 ± 1.6^b	9.1 ± 1.9^{a,b}	0.09
Deep roots	0.37 ± 0.14	0.42 ± 0.21	0.56 ± 0.17	0.74
Tap root	27.8 ± 2.2	26.1 ± 3.1	22.8 ± 2.4	0.24
Sinker roots beneath ZRT	11.2 ± 1.9^a	18.1 ± 2.4^b	17.5 ± 2.6^b	0.06
Sinker roots beyond ZRT	0.5 ± 0.4	0.6 ± 0.3	0.4 ± 0.2	0.56
Horizontal shallow segments beyond ZRT	25.2 ± 2.4	30.0 ± 3.2	31.1 ± 2.1	0.22

Where letters in superscript differ, differences are significant between treatments using a Fisher LSD test ($P < 0.05$). In the Probability (*P*) column, *P* values (significant *P* values are in bold) are given for the comparison of all three treatments together using ANOVA (with DBH as covariate). Non-proportional data were log-transformed and proportional data were arcsine square root transformed before analysis, although mean values of raw data are given here for ease of reading

relationship existed between *F*/CSA and any other variable.

When the three planting treatments were considered separately, the best predictors of anchorage differed depending on the treatment. In plants (+) tap roots, a combination of the predictors maximum root depth and lateral root number or stump proportion were highly significant (Table 4). Significant individual predictors with a high R^2 (>0.5) included the mean number of lateral roots on the stump, stump proportion, log total root volume, intermediate and shallow root volume, log maximum root depth, log stem taper and log SRS (Table 4).

In plants (–) tap roots, again a combination of predictors was found to best predict log *F*/CSA, but the best regression was found between log stem taper and volume of deep roots (Table 4). The best individual predictors were the volume of deep roots, log DBH, the volume of shallow and intermediate roots, log stem volume and log total root volume. In cuttings, no clear relationships existed between log *F*/CSA and any other parameter, as R^2 was always lower than 0.35.

Mechanical properties of individual roots

Mean E_L in all roots was 8122 ± 163 MPa and wood density was 444.5 ± 6.5 kg/m^3. No significant differences were found between lateral and tap roots. A significant relationship existed between density and E_L in all roots ($y = 27.8_{EL} - 4581$, $R^2 = 0.57$, $P < 0.001$). No significant differences were found when either density or E_L were regressed with distance from the stem in tap roots. However, both parameters were

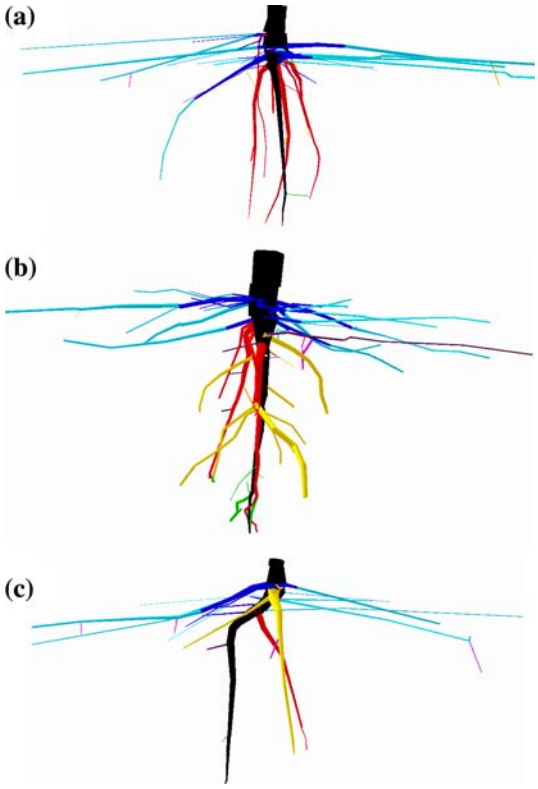


Fig. 2 Reconstruction with AMAPmod of root systems coloured as a function of compartment type in (**a**) control trees with undamaged tap roots, (**b**) trees with pruned tap roots and (**c**) cuttings. The root systems are at the same scale with a maximal rooting depth of –0.75, –0.9, and –0.8 m, respectively. Side view, left is North West

significant in lateral roots i.e. density and E_L decreased with increasing distance from the tree stem, although R^2 was low ($y = -0.956_{\text{density}} + 453.82$, $R^2 = 0.18$, $P < 0.001$; $y = -37.40_{EL} + 8204.5$, $R^2 = 0.18$, $P < 0.001$). Density increased positively with root diameter ($y = 2.36_{\text{density}} + 406.1$, $R^2 = 0.1$, $P < 0.001$) but R^2 was very low and no significant relationship was found between E_L and diameter. When regressed against log *F*/CSA, no significant relationship could be found either for all trees together or within treatments.

Mean wood density (Table 2) was significantly lower in lateral roots from cuttings and E_L was significantly lower in roots from cuttings when compared to trees (+) tap roots only (Fig. 3). No significant differences were found when roots from different quadrants were compared in any treatment. When regressed against log *F*/CSA, no significant relationship could be found either for all trees together or within treatments.

Discussion

The growth performance of seedlings was significantly better than that of cuttings, especially with regard to stem length and volume. Cuttings were always shorter than seedlings, but stem extension was similar after 6 years growth, as also found by Gaspar et al. (2005) in *Eucalyptus globulus* Labill. Seedlings (+) tap roots had the highest stem extension 3 years after transplanting and a significantly greater stem volume after 6 years. During the first few years of growth in the field, seedlings (–) tap roots and cuttings may have been allocating resources to root systems rather than to stem extension, leading to the regeneration or replacement of missing tap roots e.g. in trees where undercutting had been carried out, several new roots had regenerated from where the tap root had been cut (Fig. 2). Stem volume was the only shoot parameter which was significantly greater in plants (+) tap root compared to plants (–) tap roots, therefore undercutting had little effect on plant performance.

If any deformations in the root systems had existed due to the paperpot containers or planting out, these were probably hidden by subsequent growth (Lindström and Rune 1999). No differences in RPC occurred between treatments, therefore indicating that resource allocation to the root system was similar in all treatments after 7 years. In cuttings, total root volume was lower, which reflected the overall small size of the plants. Cuttings also had less roots and in particular fewer second order roots on the stump, as a consequence, root individual diameter and inter-lateral length on the tap root was larger. It can be supposed that when the cuttings were planted, the lateral roots which were already in position continued to thicken, with few, if any, new roots laid down. One or several roots also changed direction and grew vertically, hence giving the appearance that a tap root had formed (Fig. 2). As resources were expended to the directional growth and thickening of these roots, the %volume allocated to the oblique root compartment

Table 3 Circular distribution of (1) % root volume in several compartments (2) root length and number in the three discontinuous wind oriented sectors (WW = windward quadrant, PP perpendicular to wind direction, LW = leeward quadrant)

Variable	(+) Tap roots				(–) Tap roots				Cuttings			
	P	WW	PP	LW	*P*	WW	PP	LW	*P*	WW	PP	LW
(1) % volume in the corresponding compartments												
Total	**0.049**	27.0^{ab}	21.3^{b}	30.4^{a}	0.54	23.4	22.7	31.1	0.57	29.0	21.1	28.8
ZRT	**0.042**	29.4^{a}	19.6^{b}	31.5^{a}	0.11	19.9	24.2	31.6	**0.02**	27.4^{ab}	19.1^{b}	34.4^{a}
Horizontal beyond ZRT	0.085	28.8	19.7	31.7	0.62	18.6	24	33.4	0.14	29.0	20.1	30.7
Sinkers	0.10	20.3	22.1	35.6	0.79	27.2	22.6	27.6	**0.033**	19.7^{a}	22.0^{a}	36.3^{b}
Intermediate and oblique	0.27	20.8	23.1	33	0.63	22.5	24.2	29.1	0.63	26.8	25.9	21.5
Deep roots	0.34	29.9	17.6	34.8	0.28	33.2	17.4	32	0.37	26.0	28.3	17.5
(2) % root length and number												
Root number	0.17	27.2	22.9	27.0	**<0.001**	24.6^{a}	22.0^{a}	31.5^{b}	0.13	28.3	22.8	26.2
Root length	**0.008**	25.4^{ab}	22.0^{b}	30.6^{a}	**<0.001**	21.1^{a}	23.5^{a}	31.9^{b}	**0.042**	27.1^{ab}	21.7^{b}	29.5^{a}
Sinker root length	**0.020**	19.9	22.3	35.6	0.35	26.3	21.3	31.2	**0.013**	16.8	24.2	34.7
Root length beyond ZRT	**0.026**	26.2	20.4	33.0	**0.031**	17.3	23.9	34.9	0.75	29.0	20.0	30.9

For the PP quadrant, data used are the means of both quadrants, as it is assumed that no differences should exist between the two sectors (see Methods). Where letters in superscript differ, differences are significant between treatments using a Fisher LSD test ($P < 0.05$) and significant *P* values are in bold. ZRT = zone of rapid taper

Table 4 Regressions of *F*/CSA and tree growth parameters for treatments considered individually, along with R^2 and *P* values

Tree characteristic	Regression equation	R^2	*P*
Treatment: plants (+) tap roots			
Relative root number ($_{RRN}$) and maximum root depth ($_{depth}$)	$y = 5188 - 1488_{RRN} + 26.9_{depth}$	0.80	0.003
Mean number of lateral roots on stump ($_{lats}$) and maximum root depth ($_{depth}$)	$y = -0.44_{lats} - 1.10_{depth} + 5.45$	0.79	0.005
Stump proportion ($_{stump}$) and maximum root depth ($_{depth}$)	$y = -1.01_{stump} - 1.30_{depth} + 5.72$	0.78	0.005
Mean number of lateral roots on stump ($_{lats}$)	$y = -0.608_{lats} + 3.62$	0.69	0.003
Stump proportion ($_{stump}$)	$y = -1.44_{stump} + 3.52$	0.63	0.006
Total root volume ($_{totvol}$)	$y = -0.69_{totvol} + 3.08$	0.63	0.006
% Volume in intermediate soil layer (all root segments) ($_{intvol}$)	$y = -1.05_{intvol} + 3.56$	0.57	0.012
% Volume in shallow soil layer (all root segments) ($_{shallowvol}$)	$y = 0.937_{shallowvol} + 2.03$	0.56	0.012
Maximum root depth ($_{depth}$)	$y = 6.84_{depth} - 2.14$	0.56	0.013
Stem taper ($_{H/D}$)	$y = -0.74_{H/D} - 2.03$	0.52	0.019
Specific root surface ($_{SRS}$)	$y = 0.13_{SRS} + 0.21$	0.51	0.020
Treatment: plants (–) tap roots			
Stem taper ($_{H/D}$) and % volume in deep soil layer (all root segments) ($_{voldeep}$)	$y = 1.24_{H/D} - 4.17_{voldeep} + 0.77$	0.83	<0.001
% Volume in deep soil layer (all root segments) ($_{voldeep}$)	$y = -5.06_{voldeep} + 3.10$	0.76	<0.001
Diameter at breast height ($_{DBH}$)	$y = -1.43_{DBH} + 3.74$	0.64	0.003
Volume in shallow soil layer (all root segments) ($_{shallowvol}$)	$y = 1.63_{shallowvol} + 1.69$	0.64	0.003
Stem volume ($_{Svol}$)	$y = -2.25_{Svol} + 5.59$	0.62	0.004
Volume in intermediate soil layer (all root segments) ($_{intvol}$)	$y = -1.02_{intvol} + 3.59$	0.55	0.009
Total root volume ($_{totvol}$)	$y = -0.722_{totvol} + 3.08$	0.55	0.009
Inter-lateral length along tap root ($_{inter}$)	$y = -1.34_{inter} + 3.41$	0.50	0.015

Data is ordered with regard to significance of stem and root system variables. Data for root characteristics are proportional data for the corresponding compartments. All non-proportional data were log-transformed before analysis. All proportional (%) data were arcsine square root transformed before analysis

significantly increased (Table 2) and %volume allocation to other compartments, especially to sinker roots, decreased. Even though not the physiological tap root, this vertically growing root may serve a similar function to the tap root with regard to tree anchorage. Differences in root system characteristics between treatments (–) and (+) tap roots were also observed visually (Fig. 2)

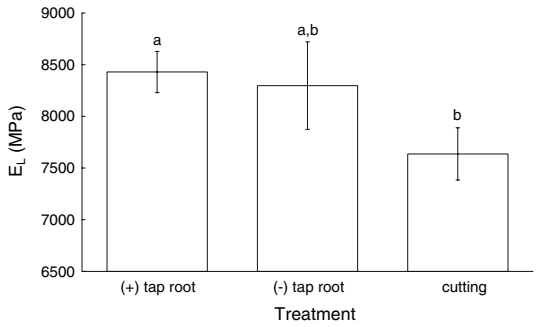


Fig. 3 Mean longitudinal modulus of elasticity (E_L) was significantly lower in roots from cuttings when compared to plants (+) tap roots only ($F_{2,46} = 2.57$, $P = 0.08$). Where letters in superscript differ, differences are significant ($P < 0.05$)

with plants (–) tap roots having one or several roots regenerating from where the tap root had been cut. However, %volume allocated to any root compartment did not differ significantly between plants (–) or (+) tap roots, even though %volume of deep roots was negatively regressed with *F*/CSA in (–) tap rooted plants. Therefore, juvenile Maritime pine has an inherently stable root system architecture, as plants which are perturbed through manipulation techniques are still able to develop root system morphologies similar to those of control plants.

The relative force necessary to deflect cuttings from the vertical was significantly less than that needed for seedlings, but when force was divided by stem CSA, no differences were found between treatments. Therefore, the lower force required to deflect the cuttings was due to differences in plant size only. These differences may become less pronounced over time, as found by Krasowski et al. (1996) in cuttings and seedlings of 11–21 year old Lodgepole pine (*Pinus contorta* Dougl. ex Loud.). We found no differences when trees were pulled in different directions, which is surprising as leeward roots were reinforced the most in terms of volume, length and number, depending on treatment. However, the trees studied were also all from within the stand and were therefore sheltered to a certain extent by the outer, edge trees. Hence, lateral loading may not have been sufficient to induce anchorage symmetry (Mickovski and Ennos 2003).

It is highly useful for forest managers to determine tree anchorage potential from growth characteristics, e.g. stem DBH, weight or volume, are frequently used as predictors of anchorage (Nicoll et al. 2006). With regard to root characteristics, few studies have used root system variables to predict root anchorage (Bailey et al. 2002; Dupuy et al. 2005a; Nicoll et al. 2006; Stokes et al. 2007), but a combination of the predictors root number and volume or size usually best predict uprooting force. When all treatments were pooled together in our study, no clear relationships were identified as R^2 was always <0.5. When treatments were considered separately, surprisingly, the best predictors of *F*/CSA differed, even though no significant differences in mean values occurred between trees. In plants (+) tap roots, a combination of maximum root depth and lateral root number or %volume in the stump were the best predictors, whereas in plants (–) tap roots, stem taper combined with %volume root segments at a depth >0.62 m was the best predictor. However, as *F*/CSA is considered here as representative of stress, negative relationships mean improved anchorage and positive relationships the inverse (Table 5). In cuttings, where no predictors of root anchorage were found and as in plants (–) tap roots, the root system was manipulated through nursery techniques, the inherent anchorage mechanisms have been altered at an early stage of development. However, after 7 years growth, little disparity in mean growth variables exist between the two populations of seedlings, although small differences, possibly due to root system asymmetry, still affect the relationships between *F*/CSA and plant growth traits. However, after several more years' growth, it is unlikely that these differences will still be noticeable (Hobbs et al. 1987; Nörr 2003).

Results from our study are similar to those found by Sasse and Sands (1997) and Watson and Tombleson (2002), in that bare-root cuttings were less developed than seedlings in 1 year old *Eucalyptus globulus* Labill. and 3 year old radiata pine (*Pinus radiata* D. Don), respectively. Watson and Tombleson (2002) compared root architecture of direct-sown and bare-root seedlings, with that of bare-root cuttings. These authors suggest that the observed wind-firmness of bare-root seedlings

Table 5 An increase in the growth of certain plant characteristics improves or decreases anchorage in plants (+) and (–) tap roots

Plants (+) tap roots	Plants (–) tap roots
Anchorage is improved when an increase occurs in the following characteristics	
Root number on stump	Volume in deep soil layer of all root segments
Stump proportion	DBH
Total root volume	Total root volume
% Volume in intermediate soil layer of all root segments	% Volume in intermediate soil layer of all root segments
Stem taper	Stem volume
Volume in windward and leeward roots	Inter-lateral length of roots on tap root
Root length on leeward side	Root number on leeward side
	Root length on leeward side
Anchorage is worse when an increase occurs in the following characteristics	
% Volume in shallow soil layer of all root segments	% Volume in shallow soil layer of all root segments
Maximum root depth	
Specific root surface	

was due to an increase in biomass in the near-stem (ZRT) lateral roots, particularly in the leeward roots. In our study, we found that total root volume and the %volume allocated to all root segments in the intermediate soil layer (hence mainly composed of tap roots and sinkers) in seedlings were good indicators of wind-firmness, whereas an increase in the %volume allocated to all root segments in the shallow soil layer suggested poor stability (Table 5). Also in agreement with Watson and Tombleson, (2002, 2004), we show that trees respond typically to prevailing winds through structural reinforcement of those parts of the root system placed under most mechanical stress (Stokes et al. 1995; Tamasi et al. 2005; Danjon et al. 2005). Our results show that the underlying mechanisms governing adaptation to wind loading were generally similar in all treatments, but in contrast with adult Maritime pine trees, saplings exhibited a selective reinforcement in windward and leeward root length, both within the ZRT and beyond, and also in leeward sinker root length. In adult trees, root length increases on the windward side of shallow roots beyond the ZRT only (Danjon et al. 2005). In both saplings and mature trees, intermediate and oblique roots and deep roots did not exhibit any asymmetry. We suggest that in our saplings, the tap root plays a major mechanical role, and that the best way to aid the tap root to maintain its position is to reinforce the ZRT and sinkers by augmenting root size and number in both leeward and windward compartments. The stump, being the upper part of the tap root is therefore also more important than in older trees without long tap roots (Nicoll and Ray 1996). Similarly, the mechanical role of roots beyond ZRT is thus less important in young trees compared to mature trees.

Cuttings were slightly straighter than seedlings, but this difference was not significant and was probably only due to the smaller size of these plants. Cuttings will have a smaller crown biomass, thus reducing crown drag, but high stem taper also increases resistance to wind loading and stem taper was significantly lower in cuttings compared to other trees. However, the shorter saplings could also have been protected from the prevailing wind by the dominant seedlings in the block. Chaperon et al. (1991) observed that cuttings of Maritime pine were straighter than seedlings 2 years after planting, but that this effect was not noticeable after 5 years of age. Lateral roots of cuttings also had significantly lower wood E_L and density compared to roots from seedlings, suggesting that this mode of reproduction can alter certain aspects of morphogenesis. A lower E_L would increase tree flexibility, thus enabling trees to bend more easily during wind loading and then to return to an upright position. The decrease of mechanical properties along lateral roots is linked to changes in xylem structure (Stokes and Guitard 1997). However, few data exist concerning this relationship in roots and more studies are needed to better understand this phenomenon.

Although bare root cuttings of certain species e.g. *Quercus rubra* L. (Schultz and Thompson 1997) and *E. globulus* (Sasse and Sands 1997), are successfully used in tree improvement programs, it is unlikely that Maritime pine cuttings will be used on a large scale, due to their poor performance with regard to stem volume, compared to seedlings the same age (Chaperon et al. 1991). The use of cuttings to determine the role of the tap root in tree anchorage is not a useful system, as one or several of the lateral roots in cuttings changes direction to ''become'' a tap root. However, this regeneration demonstrates the inherent stability of root system architectural patterns in juvenile Maritime pine, even in perturbed plants. The rigid tap root appears to play a major mechanical role in anchorage of seedlings, and its position is reinforced by lateral roots holding it in place like guy ropes (Ennos 1994). Nevertheless, questions still remain concerning the application of these results to the mature tree, where a large root-soil plate is also present. The use of numerical modelling (Dupuy et al. 2005a, b) may be the only viable, short-term method available to compare the overturning resistance of adult trees in similar conditions with the same root architecture, but with/without presence of any given root. Similarly, further work needs carrying out on how soil properties can be used to predict subsequent uprooting force (Moore 2000; Nicoll et al. 2006). In particular, soil moisture content and compaction can significantly alter both anchorage resistance and the mode of failure during loading (Ennos 1990; Goodman and Ennos 1999). Nevertheless, our study has showed that the most useful indicators of root anchorage depend on how seedlings have been treated in the nursery and the subsequent transplantation techniques, which will probably influence tree mechanical stability later on in life.

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